

JOSHUA PATTERSON

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Software engineer with 3+ years experience building full-stack systems for federally funded research. Specializes in API design, data pipelines, and workflow automation; experienced leading development and mentoring junior engineers.

WORK EXPERIENCE

Research Associate

Aug. 2023-Present

Florida Institute for Human & Machine Cognition | Pensacola, FL

- Lead architecture and development of internal software systems for federally funded research, automating experimental workflows and data processing pipelines.
- Design and implement RESTful APIs and WebSocket services using Python (FastAPI, Flask) to enable real-time communication and automation across distributed research applications.
- Build data pipelines for large-scale time-series datasets using Pandas and SciPy, processing and preparing experimental data for analysis and visualization.
- Architect modular, object-oriented systems using Pydantic models and interface-driven design, enabling consistent patterns across multiple concurrent research projects.
- Integrate relational and document databases (PostgreSQL, MySQL, MongoDB, CouchDB, SQLite) into research applications, selecting storage strategies based on data structure and access
- Mentor software development interns through onboarding, code review, and task delegation; maintain code quality and architectural consistency across contributions.

Research Intern

Feb. 2023-Aug. 2023

Florida Institute for Human & Machine Cognition | Pensacola, FL

- Automated system configuration and validation processes using Python and Bash scripting, reducing manual setup time and improving system reliability for experimental environments.
- Built internal dashboards and visualization tools using Plotly Dash to support data exploration, validation, and stakeholder insight generation.

Undergraduate Research Assistant

Aug. 2022-May 2023

University of West Florida Center for Cybersecurity and AI | Pensacola, FL

- Developed web interfaces for an OSINT vehicle-security portal using C# and JavaScript, including a CVSS calculator to standardize vulnerability scoring and support security assessments.

Software Engineer I - Intern

May 2022-Aug. 2022

Dynamic Software Solutions (DS2) | Niceville, FL

- Developed software components using C#, SQL Server, and game engines (Unity, Unreal Engine 5), contributing to interactive and simulation-based applications.

EDUCATION

Master of Science in Cybersecurity / Software & System Security

May 2025

University of West Florida

Bachelor of Science in Software Design and Development, Minor in Computer Science

May 2023

University of West Florida

- Certificates in Cybersecurity and Database Systems

CERTIFICATIONS

CompTIA Security+

Nov. 2025

CITI Program Export Compliance (EC)

Aug. 2025

CITI Program Biomedical Research

Mar. 2023

SKILLS

Languages: Python, Java, JavaScript, C#

Frameworks/Tools: FastAPI, Flask, Node.js, Plotly Dash, Pandas, NumPy, SciPy, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, CouchDB, SQLite

Technologies: REST APIs, Microservices, Data Pipelines, Bash, Git, Docker, AWS